Donghee (Joy) Chae

Personal website | Youtube | donghee7878@gmail.com | chaedh1213@korea.ac.kr

Research Interests

To engineer immune-instructive materials for cancer and inflammatory disease, and to keep them intelligible to the public they are built for.

- **Stem cell-based immunomodulation.** MSC immunosuppression and efferocytosis-driven reprogramming of monocytes and macrophages.
- **Nanomaterials.** Radiation-responsive scintillating nanoparticles for X-ray-triggered uncaging; non-viral nanocomplexes for CAR-TAM.
- **Bioinformatics and *in silico* modeling.** HER2-ADC response prediction, tumor immune remodeling, and structure-based drug repurposing.
- **Environmental exposure and immune function.** Pb, PCBs, and PFAS as modifiers of antigen presentation and cellular immunity.

Education

Korea University | Seoul, Republic of Korea — **Mar 2022 – Feb 2027** *(expected)*
B.S. in Health and Environmental Science
B.E. in Biomedical Engineering (dual degree, admitted Nov 2023)

University of Iowa | Iowa City, IA, USA — **Aug 2025 – May 2026**
Exchange Student, Roy J. Carver Department of Biomedical Engineering

Research Experience

Nano-Biomedicine Lab, Korea University (PI: Mikyung Kang) — *Seoul, Korea*
Undergraduate Researcher — **Jun 2026 – Present**

- **Radiation-activated immunotherapy** (KIRAMS and Harvard Medical School collaboration)
  - Synthesized **inorganic nanoparticles** that convert X-ray energy into UV emission to cleave a UV-labile linker.
  - Designed the **in silico drug repurposing** workflow used to identify an IL-2-mimetic docking system.
- **Propagation of MSC-induced immunosuppression to macrophages**
  - Tested whether the immunosuppressive phenotype of **efferocytic macrophages** propagates to **bystander macrophages** through exosome-mediated transfer.
- **HER2 ADC response prediction** (KIST molecular imaging collaboration)
  - Analyzed GEO datasets to prioritize biomarkers of T-DXd and T-DM1 response.
  - Tracked antibody internalization and **lysosomal degradation** by immunocytochemistry and confocal imaging.
  - Built a **predictive mathematical model** linking trafficking-gene expression to ADC response.
- **CAR-TAM drug factory** (Korea University, Kyung Hee University, and Yale Cancer Center collaboration)
  - Developed the non-viral delivery system replacing lentiviral vectors, using PEI-plasmid nanocomplexes that deliver CRISPR-Cas9 and an HITI donor for single-step SIGLEC10 knockout and CAR knock-in.
- **Environmental immunotoxicology**
  - Designed a study asking whether non-cytotoxic **Pb** exposure impairs **phagolysosomal** antigen processing and antigen-specific $CD8^+$ T-cell activation in DC2.4 dendritic cells.

Ankrum Lab, University of Iowa (PI: James Ankrum) — *Iowa City, IA*
Undergraduate Research Intern — **Aug 2025 – May 2026**

- **Efferocytosis of cytokine pre-licensing of MSCs**
  - Showed that **cytokine pre-licensing of MSCs** elicits the efferocytic response of primary human monocytes, with downstream **suppression of T-cell proliferation**; led manuscript preparation.
  - Validated the five-color monocyte/macrophage phenotyping panel (CD14, CD16, CD86, CD163, CD206) on flow cytometry and performed all FlowJo analysis.
  - Quantified secreted TNF-α, IFN-γ, IL-10, and IL-4 using ELISA.
- **Subacute PCB inhalation toxicology** (with Prof. Andrea Adamcakova-Dodd)
  - Ran the murine inhalation exposure study on PCB1254 combined with a Western diet.
  - Performed mice necropsy and multi-organ harvest, followed by nucleic acid extraction and PCR.
- **Single-cell energy metabolism of islet alpha and beta cells** (SCENITH, with **Prof. Yumi Imai**)
  - Analyzed SCENITH flow cytometry measuring **puromycin incorporation** as an index of ATP production in intact mouse islets under glucose and fatty acid deprivation.
  - Resolved alpha cells and two beta cell subpopulations by puromycin signal and confirmed them as physically distinct populations by back-gating onto FSC/SSC.

ChEEHR Lab, Korea University (PI: Yoon-Hyeong Choi) — *Seoul, Korea*
Undergraduate Researcher — **Jun 2024 –Aug 2024**

- Analyzed post-disaster exposure data from the **Humidifier Disinfectant Incident** for national health risk assessment.
- Curated heavy metal exposure data and performed the epidemiological analysis.

Publications

Accepted

[1] K. Adajar[†], S. Liu[†], F. Haider, **D. Chae**, R. J. Byrd, O. A. Louis, J. A. Ankrum, and Y. Imai[*], *"Lipolysis supports glucagon secretion from human and mouse islets independent of ATP production in alpha cells assessed by SCENITH,"* ***Frontiers in Endocrinology***, accepted (2026).

In preparation

[2] **D. Chae**, D. del Bosque Siller, J. N. Liszewski, R. Byrd, M. V. Schrodt, R. M. Behan-Bush, and J. A. Ankrum[*], *"Cytokine pre-licensing of mesenchymal stromal cells amplifies the efferocytic immunosuppressive response of monocytes and their macrophage progeny,"* in preparation.

[3] **D. Chae**, M. Kim, Y. Y. Hwang, C. You, W. Ju, I. Noh[*], and M. Kang[*], *"Beyond ex vivo CAR-T therapies through in vivo programming and targeted delivery,"* in preparation.

[4] J. Jeong[†], **D. Chae**[†], J. Lee, S. Park, M. A. Miller, and M. Kang[*], *"Radiation-responsive nanomaterials: mechanistic foundations and emerging frontiers in cancer theranostics,"* in preparation.

[5] A. Adamcakova-Dodd, H.-J. Lehmler, X. Jing, H. Wang, A. Klingelhutz, J. A. Ankrum, **D. Chae**, D. May, and P. S. Thorne[*], *"Subacute inhalation exposure to PCB1254 with a Western diet alters metabolic and pulmonary inflammatory responses in mice,"* in preparation.

[†]Equal contribution, [*]Corresponding author

Conference Presentations & Invited Talks

[1] **D. Chae** & M. Kang et al., *"Efferocytosis of mesenchymal stromal cells potentiates an immunosuppressive monocyte phenotype that endures after macrophage differentiation,"* **KSBM 2026** (Korean Society for Biomaterials Annual Meeting), Busan, Korea. (2026, September).

[2] S. Liu, F. Haider, **D. Chae**, L. Kim, R. Byrd, J. A. Ankrum, Yumi Imai[*], *"A role of alpha cell fatty acid metabolism in glucagon secretion,"* **Midwest Islet Club 2026**, Madison, WI. (2026, May).

[3] **D. Chae**, *"Cytokine pre-licensing of MSCs elicit an enhanced efferocytic response from monocytes leading to improved T-cell suppression,"* **Research Open House 2026**, College of Engineering, University of Iowa, Iowa City, IA. (2026, April).

[4] **D. Chae** & J. A. Ankrum et al., *"Cytokine pre-licensed MSCs elicit an enhanced efferocytic response from monocytes leading to improved T-cell suppression,"* **ISCT 2026 Annual Meeting**, Dublin, Ireland. (2026, February). Abstract published in ***Cytotherapy***.

[5] **D. Chae** and H. Yang, *"Non-viral intron knock-ins unlock splicing-controlled CAR-T cell therapies,"* **K-BioX Summit 2025**, Seoul, Korea. (2025, May). Advisor: **Prof. Theodore Roth**, Stanford University.

Invited Talks

[6] **D. Chae**, Senior Learning Strategy Lecture, 2026 Learning Support Lecture and Workshop Series, Center for Teaching and Learning, Korea University, Seoul, Korea. (2026, October). Invited lecture; selected as an outstanding graduating student.

[7] **D. Chae**, Korea–U.S. High-Tech Industry Youth Exchange Program Annual Seminar, KIAT, Seoul, Korea. (2026, July). Invited reflection talk.

Honors & Awards

President's Award, Korea Institute for Advancement of Technology (KIAT) — **May 2026**
Most outstanding student of the Korea–U.S. High-Tech Industry Youth Exchange Support Program.

Minister's Award, Ministry of Trade, Industry and Energy (MOTIE) — **Jan 2026**
For science communication as a member of the 3rd KORUS 70 Press Corps.

Excellence Award, ResearchThon, AI BioX ConfEX Grand Summit | PHNYX × Cheiron × K-BioX — **Jan 2026**
"Development of an Adapter System to Minimize Air Contamination during Venous Blood Gas (VBG) Sampling."

Final Outstanding Team, B.I.O. Biotechnology Joint Academic Society | Korea University × Yonsei University — **Jun 2025**

Grand Award (Best Poster Presentation), K-BioX Summit 2025 | Yonsei University × Rutgers University × HTAN × G1's Lab — **May 2025**

Outstanding Team, COCF, National Climate Initiative Competition — **Sep 2024**

Skills & Techniques

Cell culture	ucMSC, PBMC, Monocyte isolation, THP-1 and THP-1-derived macrophages, DC2.4, NR8383, iPSC, adipocyte
Functional assays	Efferocytosis assays, T-cell suppression (CFSE), macrophage polarization and phenotyping, LDH/CCK-8 viability assays
Immunoassays	Flow cytometry and FlowJo, ELISA, SDS-PAGE, Western blot, Immunocytochemistry, Bradford/BCA assays
Molecular biology	Nucleic acid extraction, RT-PCR and qPCR, primer design, plasmid preparation, electroporation
In vivo	Murine inhalation exposure system operation, necropsy and multi-organ harvest, brain tissue dissection, brain cell isolation
Nanomaterials	Core/shell lanthanide scintillating nanoparticles ($NaLuF_4$:Dy/Tm/Nd,Gd@$NaYF_4$, LaF_3:Ce^{3+}/LaF_3-OA), mesoporous silica and iron oxide nanoparticles, LNP and PEI polyplex formulation, DLS/ELS, FT-IR, NanoDrop, ferrozine assay
Imaging	Confocal microscopy, fluorescence live-cell imaging with membrane and organelle trackers, lipid staining
Data Analysis	R (Bioinformatics; bulk RNA-seq, GEO analysis, gene module construction), MATLAB, SPSS *ADsP (Advanced Data Analytics Semi-Professional) certified, Korea Data Agency (2025, March).
In silico	Structure-based drug repurposing: AlphaFold 3, hotspot identification, AutoDock Vina virtual screening
Visualization	BioRender, Rhino 3D, ImageJ, Affinity.

Teaching & Extracurricular Activities

Volunteer | World Vision, CHA Gangnam Medical Center, IHCO — **Jan 2024 – Present**
Over 120 hours of English–Korean translation for sponsored children's correspondence and workshop leadership for new volunteers; pediatric and emergency department support and monthly rural mobile clinic visits.

Academic Consultant | Modeun Education — **Dec 2023 – Present**
Advise high school students on research-based academic portfolios, inquiry report design, personal statements, and admissions interviews; design one-to-one curricula.

Press Corps Member (3rd Cohort, selected nationally) | KORUS 70 / MOTIE — **Jul 2025 – May 2026**
Produced video and wrote columns on Korea–U.S. STEM research exchange for a general audience.

Global Ambassador, SIMTOMI | Mediark, medical AI healthcare startup — **Aug 2025 – Oct 2025**
Promoted a medical AI application through video content and marketing campaigns built around narrowing gaps in access to care; represented the platform to international student and researcher communities.

Project Coordinator, B.I.O. Joint Academic Society | Korea University and Yonsei University — **Feb 2025 – Jun 2025**
Led team BMN2 through an interdisciplinary immunology literature review and discussion project.

Vice President, Environmental Health Science (EHS) Society | Korea University — **Mar 2024 – Jan 2025**
Directed the society across the three fields of industrial health and safety, organizing public health campaigns and campus-wide risk assessment projects.

Lead Guitarist and Organizing Committee | TRUSS, Korea University student rock band — **Mar 2023 – Jan 2025**
Performed electric guitar in six live shows over two years; organizing committee member for rehearsal scheduling, venue logistics, and stage production.

Mathematics Instructor | Megastudy — **May 2022 – Dec 2024**
Taught over 200+ students across 10 schools; developed lesson plans, lecture materials, and visual content for the national e-learning platform.

Climate Policy Researcher, Climate Scouts (9th Cohort) | COCF, non-profit foundation — **May 2024 – Sep 2024**
Delivered carbon-reduction education to elementary and middle school students, and enacted a carbon credit policy proposal to the Ministry of Environment.

Scholarships

STEM Talent Growth Pathway Program | Ministry of Education / National Research Foundation of Korea — **Sep 2026 – Feb 2027**
Undergraduate research scholarship of KRW 1,000,000 (~USD 700) per month

Udeok Foundation Scholarship | Hanil Cement — **Mar 2024 – Graduation**
Merit- and need-based award of KRW 4,500,000 (~USD 3,200) per semester over four semesters (USD 13,000 total)

Korea–U.S. High-Tech Industry Youth Exchange Scholarship (3rd Cohort) | KIAT / MOTIE — **May 2025 – May 2026**
Selected on academic merit and potential contribution to Korea's STEM industry (USD 9,000)

Languages

Korean (Native Speaker) | **English** (TOEFL 5.0/6.0 equivalent to 100/120) | **Chinese** (HSK Level 2)

References

James Ankrum, Ph.D. · james-ankrum@uiowa.edu
Professor, Roy J. Carver Dept. of Biomedical Engineering, University of Iowa

Andrea Adamcakova-Dodd, Ph.D. · andrea-a-dodd@uiowa.edu
Adjunct Assistant Professor, College of Public Health, University of Iowa

Yumi Imai, M.D. · yumi-imai@uiowa.edu
Professor, Endocrinology and Metabolism, University of Iowa

Theodore Roth, M.D., Ph.D. · troth@stanford.edu
Assistant Professor, Department of Pathology, Stanford Medicine

Yoon-Hyeong Choi, Ph.D. · yoonchoi@korea.ac.kr
Professor, Health and Environmental Science, Korea University

Mikyung Kang, Ph.D. · ieebun@korea.ac.kr
Associate Professor, Health and Environmental Science, Korea University

Aloysius Klingelhutz, Ph.D. · al-klingelhutz@uiowa.edu
Professor, Microbiology and Immunology, University of Iowa

Siming Liu, Ph.D. · siming-liu@uiowa.edu
Imai Lab, Department of Internal Medicine, University of Iowa

Aram Chung, Ph.D. · ac467@korea.ac.kr
Professor, School of Biomedical Engineering, Korea University